

Research Article

Proton irradiation assisted localized corrosion and stress corrosion cracking in 304 nuclear grade stainless steel in simulated primary PWR water

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ARTICLE INFO

Article history:

Received 16 December 2019

Received in revised form 19 March 2020

Accepted 3 April 2020

Available online 12 July 2020

Keywords:

Stainless steel

AFM

TEM

High temperature corrosion

Stress corrosion

ABSTRACT

Localized deformation and corrosion in irradiated 304 nuclear grade stainless steel in simulated primary water were investigated. The investigation was conducted by comparing the deformation structure, the oxide scale formed at the deformation structure, and their correlation with cracking. The results revealed that increasing the irradiation dose promoted localized corrosion at the slip step and grain boundary, which was primarily attributed to the strain concentration induced by enhanced localized deformation and depletion of Cr at grain boundary. Further, a synergic effect of the enhanced localized deformation and localized corrosion at the slip step and grain boundary caused a higher cracking susceptibility of the irradiated steel.

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1. Introduction

The austenitic stainless steel (SS) is widely used in the primary circuit of pressurized water reactors (PWRs) for fabrication of core structural components. The steel, though typically corrosion resistant, was found by both laboratory experiments and plant experiences to be susceptible to irradiation-assisted stress corrosion cracking (IASCC) in primary PWR water [1–4]. Due to the radioactivity of neutrons, the effect of irradiation on materials is usually studied by using simulation irradiation such as protons and $\text{Fe}^{3+}/\text{Fe}^{2+}$. To date, a number of studies have been carried out to clarify the mechanism of IASCC of the austenitic SS [5–15]. Lear et al. compared the IASCC behavior of 13 alloys in two environments and suggested that the irradiation microstructure and irradiation hardening are very necessary for IASCC [14]. Miura et al. focused on the role of grain orientation on cracking formation in irradiated SS, which indicated pure mechanical strength can lead to cracking [15]. The deformation structures in the irradiated steel were

usually observed as surface slip steps because of formation of dislocation channels in the subsurface, in which the radiation-induced defects, such as dislocations and precipitates, can mitigate dislocation glide. As such the deformation was localized into the channel, which can be a potential contributor to IASCC by promoting dislocation pile-ups at the grain boundary region [6–8]. Further, the dislocation channeling may also cause a rupture of the oxide film before the base metal cracking by promoting localized grain boundary deformation, which further leads to exposure of the metal to the high temperature water and subsequent corrosion and IASCC [7,16–19]. These investigations indicated that the change in deformation modes and the interaction of dislocation channels with the grain boundary were key factors promoting the IASCC.

Similar to stress corrosion cracking (SCC), IASCC is also a synergic effect of an aggressive environment, a susceptible material and the applied stress/strain. Since corrosion is one of the key factors affecting the SCC [20,21], a better understanding of the role of corrosion in IASCC is needed in addition to the role of the localized deformation. This requires at first to understand whether and how the irradiation affects the corrosion. In fact, the promotional effect of irradiation on corrosion has been confirmed [22–24]. It was found that irradiation promoted both the overall and intergranular corrosion of austenitic SS in primary PWR water, which was attributed to the radiation induced defects as well as the Cr

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Table 1
Chemical composition (wt.%) of 304NG SS.

C	Mn	Si	S	P	Ni	Cr	Co	Fe
0.04	1.73	0.27	0.002	0.021	8.87	19.51	0.04	Bal.

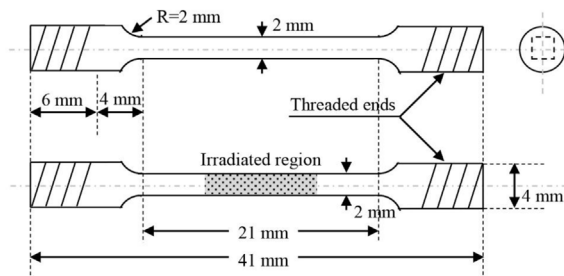


Fig. 1. Geometry and dimension of the tensile specimen.

depletion at the grain boundary [22]. Secondly, as reported by literatures [25–28], deformation could result in formation of areas with a high density of dislocations and accumulation of localized strain at the deformation area. Such changes promote corrosion of austenitic alloys in high temperature water by the enhanced diffusion of species through the dislocation [25,29,30]. Since the IASCC usually initiates locally at grain boundary regions with which the slip steps interact [5–7,17–19], the localized corrosion at the grain boundary region may also be affected by the localized deformation. This suggests that to fully clarify the mechanism of the promotion effect of localized deformation on IASCC, the synergic effect between the corrosion and localized deformation needs to be clarified, in particular to the corrosion behavior at such deformation regions as the slip step and grain boundary. To date, the corrosion at slip step and grain boundary as well as their correlation with IASCC has not received much attention. While one study concluded that no penetrative oxidation was observed along the slip step induced by pre-straining in air, the synergic effect between synchronized deformation and corrosion was not considered [31]. In the actual process of SCC, corrosion occurs simultaneously with localized deformation in the vicinity of the crack tip [32,33], suggesting that the promoted corrosion at the slip step may occur in such a case.

This study focused on the synergic effect of localized deformation and corrosion in irradiated 304 nuclear grade (NG) SS in simulated primary water of PWR. The objective is to correlate the localized deformation to the corrosion at the slip step and grain boundary, as well as the localized deformation and corrosion to IASCC. The specimens irradiated to various doses by proton were strained by constant extension rate tensile (CERT) at a fixed strain rate of 3×10^{-7} /s in simulated primary PWR water, in efforts to investigate the localized corrosion and IASCC behavior during straining. Following the CERT test, characterization of the localized deformation, corrosion and IASCC behaviors was conducted by using atomic force microscopy (AFM), electron backscatter diffraction (EBSD), transmission electron microscope (TEM) and scanning electron microscope (SEM).

2. Experimental

2.1. Material and specimens

The material used was solution annealed (SA) 304NG SS with a chemical composition listed in Table 1. Average grain size of the material was in the range of about 2050 μm . Small-sized tensile specimens with the geometry and dimension shown in Fig. 1 were fabricated from the steel and used for the IASCC tests. Each

specimen with two threaded ends had a 2 mm $\times$ 2 mm squared gauge section with a gauge length of 21 mm. The specimen was wet ground using SiC papers up to 4000 grit, followed by 1- μm diamond paste polishing to obtain a mirror surface. Then the specimen was exposed to 40-nm colloidal silica slurry polishing for over 2 h to produce a surface free of residual strain prior to proton irradiation. At last, all specimens were irradiated by protons of 2 MeV at 360 $^{\circ}\text{C}$ using the 3 MV pelletron accelerator at the Michigan Ion Beam Laboratory, to a dose of 0.5, 1.5 and 5 displacement per atom (dpa), respectively. It is worth noting that the dose here was calculated with full cascade option. The proton beam was raster-scanned over a 10 mm length in the gauge section of the specimens, creating a nearly uniform damage profile within the ~ 1 to ~ 10 μm of the whole 20 μm -thick damaged layer [22]. The dose rate as calculated by using the Stopping and Range of Ions in Materials 2008 (SRIM 2008) computer code was about 5.96×10^6 dpa per second (dpa/s). Detailed description of the irradiation process is available elsewhere [6,34].

Following the irradiation, the irradiated specimens were polished again by 40-nm colloidal silica slurry for about 1 h to get fresh surfaces, which removed the surface with a thickness of 12 μm and could ensure that the specimen surfaces following different doses of irradiation were consistent.

2.2. High temperature water test

A low flow rate (~ 100 mL/min) refreshed loop equipped with a 3-L autoclave made of 316 SS was used for the IASCC tests in simulated primary PWR water at 320 $^{\circ}\text{C}$ and 13 MPa. The flow and pressure of water were maintained using a commercial high-pressure pump in connection with a backpressure valve. Temperature in the autoclave was achieved by one pre-heater at the autoclave inlet and two resistance heating bands at the autoclave surface, which was controlled by the k-type thermocouple with an accuracy of ± 1 $^{\circ}\text{C}$. The primary water was prepared by mixing deionized water with 1200 mg/L of B as H_3BO_3 and 2.3 mg/L of Li as $\text{LiOH} \cdot \text{H}_2\text{O}$. Before ramping the autoclave to target temperature, concentration of dissolved oxygen (DO) in the inlet water was controlled at less than 5 $\mu\text{g/L}$ by bubbling H_2 gas into the water tank, followed by storing H_2 in the tank with a hydrogen overpressure of 0.08 MPa, resulting in a dissolved hydrogen (DH) concentration of 2.6 mg/L. DO, DH and conductivity of the inlet water were continuously monitored during the tests.

The 0.5-, 1.5- and 5-dpa irradiated specimens as well as the SA specimen with no irradiation were employed for the IASCC tests. These specimens were strained by CERT at a fixed strain rate of 3×10^{-7} /s in the primary water using a 50 kN servo motor in the constant displacement rate mode. The servo motor was provided by Interactive Instruments Inc. The test was interrupted at the strain of 1% and 3% for all the specimens in an effort to study the effect of applied strain on IASCC initiation behavior. Prior to CERT, the specimens were exposed in the environment at a small load of approximate 50 N for a period of three days to stabilize the environmental condition. Throughout the test, the load, displacement and environmental parameters were collected and recorded in every thirty seconds using a personal computer equipped with a LabView software. At each interrupted strain, the specimens were removed from the autoclave for a characterization of the deformation and IASCC behavior on the irradiated region of the specimens. Following the interruption, the specimens were installed into the autoclave again and the test was continued to next targeted strain with the same condition as that employed before the interruption.

To correlate the localized corrosion with the IASCC, following the strain of 3%, the oxide scale formed at grain boundary and slip step in the SA and irradiated specimens was characterized to clarify the localized corrosion behavior. Detailed description of the charac-

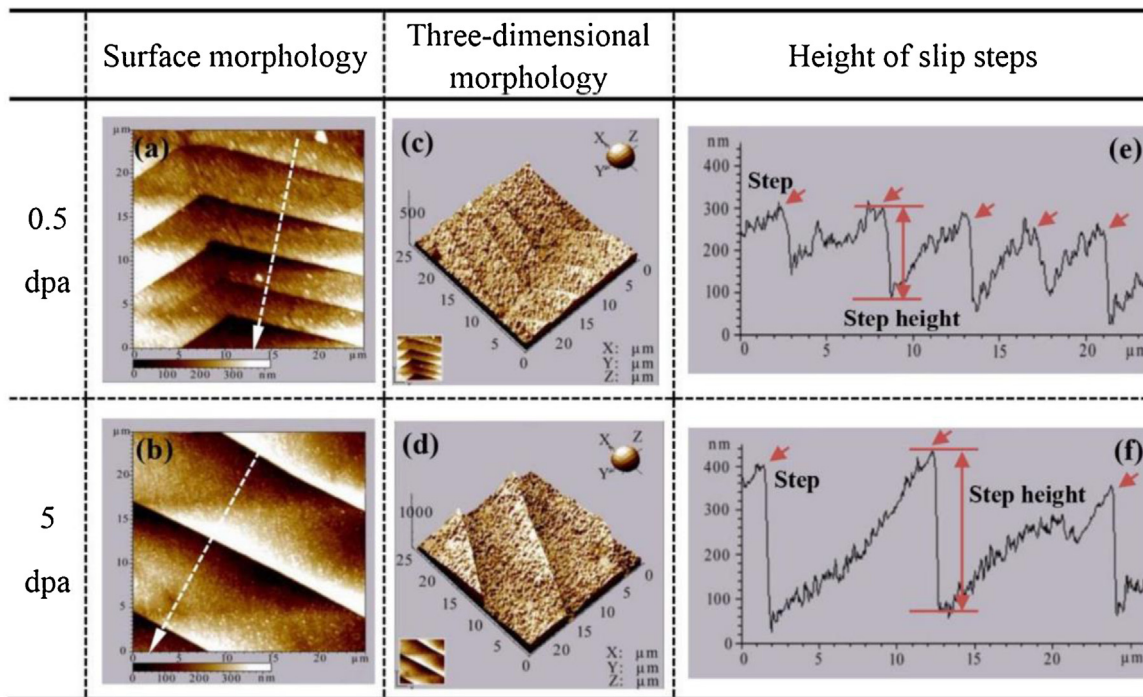


Fig. 2. AFM observation and analysis of the slip steps in 0.5- and 5-dpa irradiated specimens following 3% straining. (a) and (b) Typical surface topographic images of slip steps. (c) and (d) The corresponding three-dimensional morphology of the slip steps. (e) and (f) Surface height taken along the dashed line indicated in (a) and (b), respectively.

terization of the corrosion behavior was presented below in Section 2.4.

2.3. Characterization of the localized deformation

At each interrupted strain, characterization of the deformation behavior was conducted, including the height of slip step and localized residual strain at the slip step and grain boundary.

2.3.1. Slip step height characterized by AFM

The slip step height is determined by the number of dislocation in the dislocation channel, and it corresponds to the amount of localized deformation [7,8], so that it was selected as the quality indicator to quantify the localized deformation. Further, due to the grain orientation variation upon the tensile direction, the slip step height varied significantly from grain to grain. Therefore, the weighted slip step height itself was employed as the weight factor [6]:

$$\bar{h} = \frac{\sum_{i=1}^n h_i^2}{\sum_{i=1}^n h_i} \quad (1)$$

where $\bar{h}$ is the weighted slip step height and h_i is the height of slip step i . The height of slip step was measured using a Pico Plus AFM equipped with an EV scanner. The resolution of the scanner was less than 1 nm in the vertical direction, which is sufficient for characterizing the slip step height. As the space between slip steps was generally a few micrometers, a suitable scan area of $25 \mu\text{m} \times 25 \mu\text{m}$ was employed, covering a couple of steps for each scanning. More than 20 grains with a total of 100200 slip steps were characterized for each of the SA, 0.5-, 1.5-, and 5-dpa irradiated specimens after each interrupted strain.

2.3.2. Localized residual strain analyzed by EBSD

A FEI XL30 Field Emission Gun SEM (FEG-SEM) was used for analyzing the distributions of localized residual strain at slip step and grain boundary by EBSD in connection with the HKL-Tango soft-

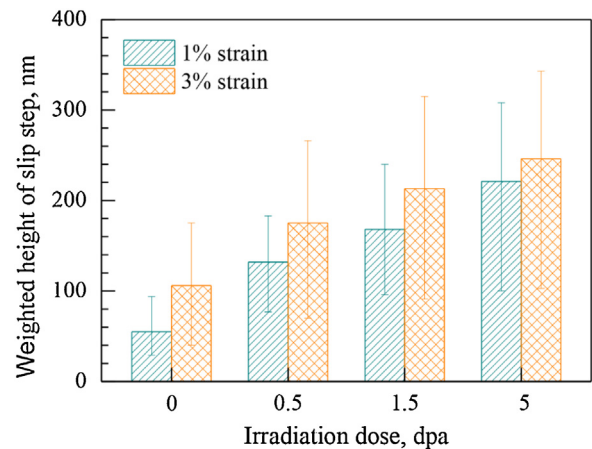


Fig. 3. Weighted average slip step height of 304NG SS as a function of irradiation dose.

ware. The scanning step size for EBSD was set as $0.45 \mu\text{m}$ using the voltage of 25 kV. The local change in the crystal orientation was evaluated by measuring the local misorientation (M_L), which denoted local strain difference by indexing and averaging the misorientations between adjacent measured points within the grains [35]. The M_L was an analogy of grain reference orientation deviation, and had a linear relationship with the degree of strain [35], which was used for describing the strain concentration at the grain boundary and slip step in the strained specimens. Details about the M_L calculation and analysis were presented elsewhere [35]. By setting the grain tolerance angle at 5° , the two neighboring parts showing a misorientation within the tolerance value were considered to be in the same grain. The 0.5- and 5-dpa irradiated specimens following the strain of 3% in the simulated primary PWR water were selected for EBSD analysis. Prior to the analysis, the specimens were electropolished in an electrolyte (10 vol.% perchloric acid of 70 wt.% concentration and 90 vol.% glacial acetic acid of

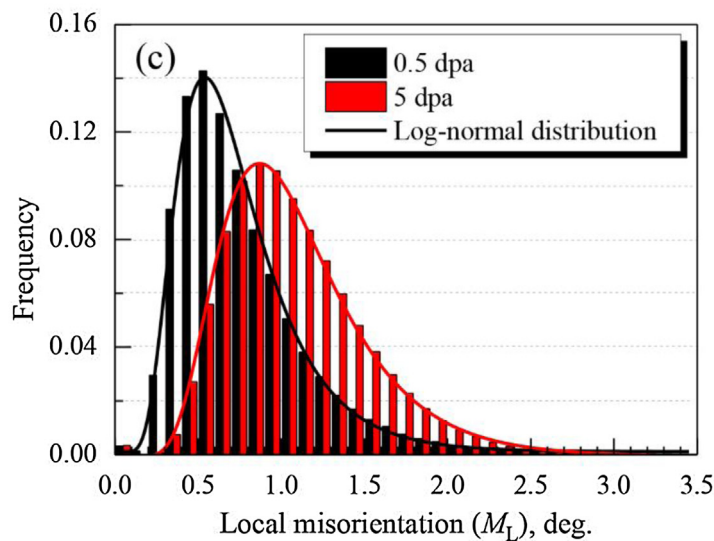
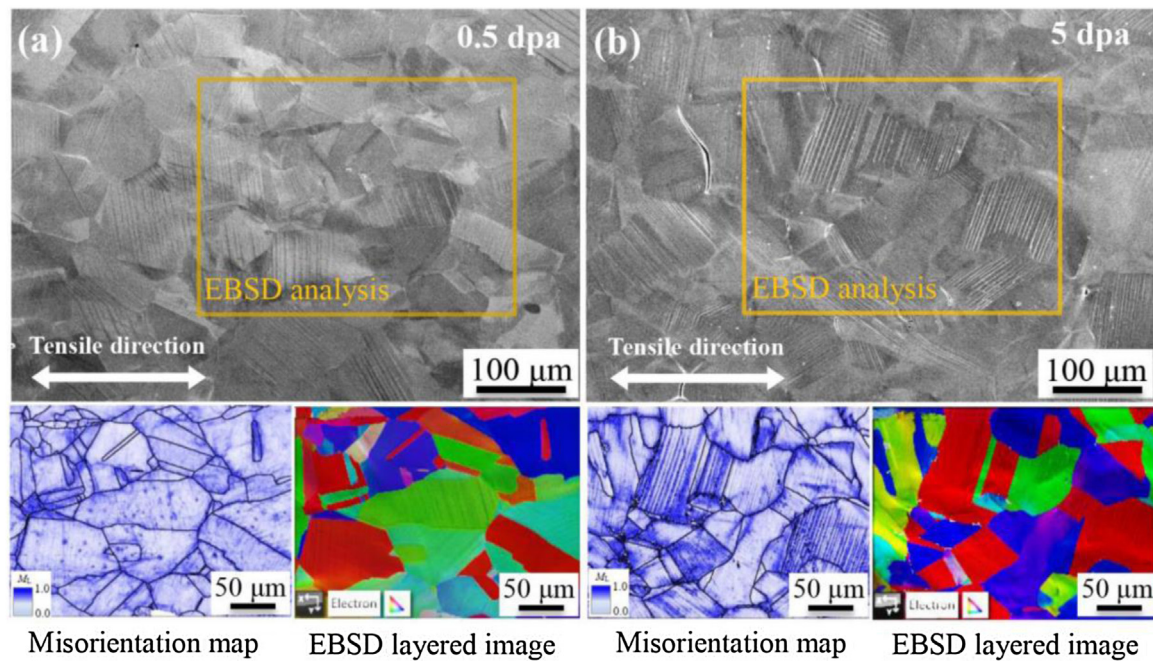


Fig. 4. Typical SEM images and EBSD analysis for 0.5- (a) and 5-dpa (b) irradiated specimens following 3% straining in simulated primary PWR water at 320 °C. (c), distribution of local misorientations obtained from (a) and (b).

98 wt.% concentration) at $-10\text{ }^{\circ}\text{C}$ and 20 V for 10 s to remove the surface oxide scale and surface residual strain.

2.4. Analyses of the corrosion behavior

Following the strain of 3%, cross-sections of the oxide scale formed on the specimens were characterized using a JEM-2100 TEM. The TEM was equipped with a Gatan 4k charge coupled device camera for capturing high-resolution TEM images, and an energy dispersive X-ray spectroscopy (EDX) device for the composition analysis using both point- and line-scans. The high-resolution TEM images collected were analyzed using the Fast Fourier Transform (FFT) technique for crystallographic details of the oxide scale. The TEM samples were prepared using a dual-beam focus ion beam (FIB)-SEM technique with a Quanta 200 3D system, so that precise sampling of the cross-section of a selected slip step and/or grain

boundary can be attained, allowing the observation and analysis of the corrosion at the slip step and grain boundary. The high-angle grain boundaries (HAGBs) were of interest to select, which ensured by using ion channeling contrast of the grains [36]. Before the TEM analysis, the selected grain boundaries were identified again by the electron diffraction pattern of TEM to ensure that they were HAGBs. A total of thirteen grain boundaries and thirteen slip steps were analyzed, including three grain boundaries for the SA specimens, four grain boundaries and six slip steps for the 0.5-dpa irradiated specimens, and six grain boundaries and seven slip steps for the 5-dpa irradiated specimens. In addition, depth of the oxide formed at the grain boundary in strained and strain-free specimens was compared to correlate localized deformation to the intergranular corrosion. The data of the oxide depth for strain-free specimens was available in a previous study [22].

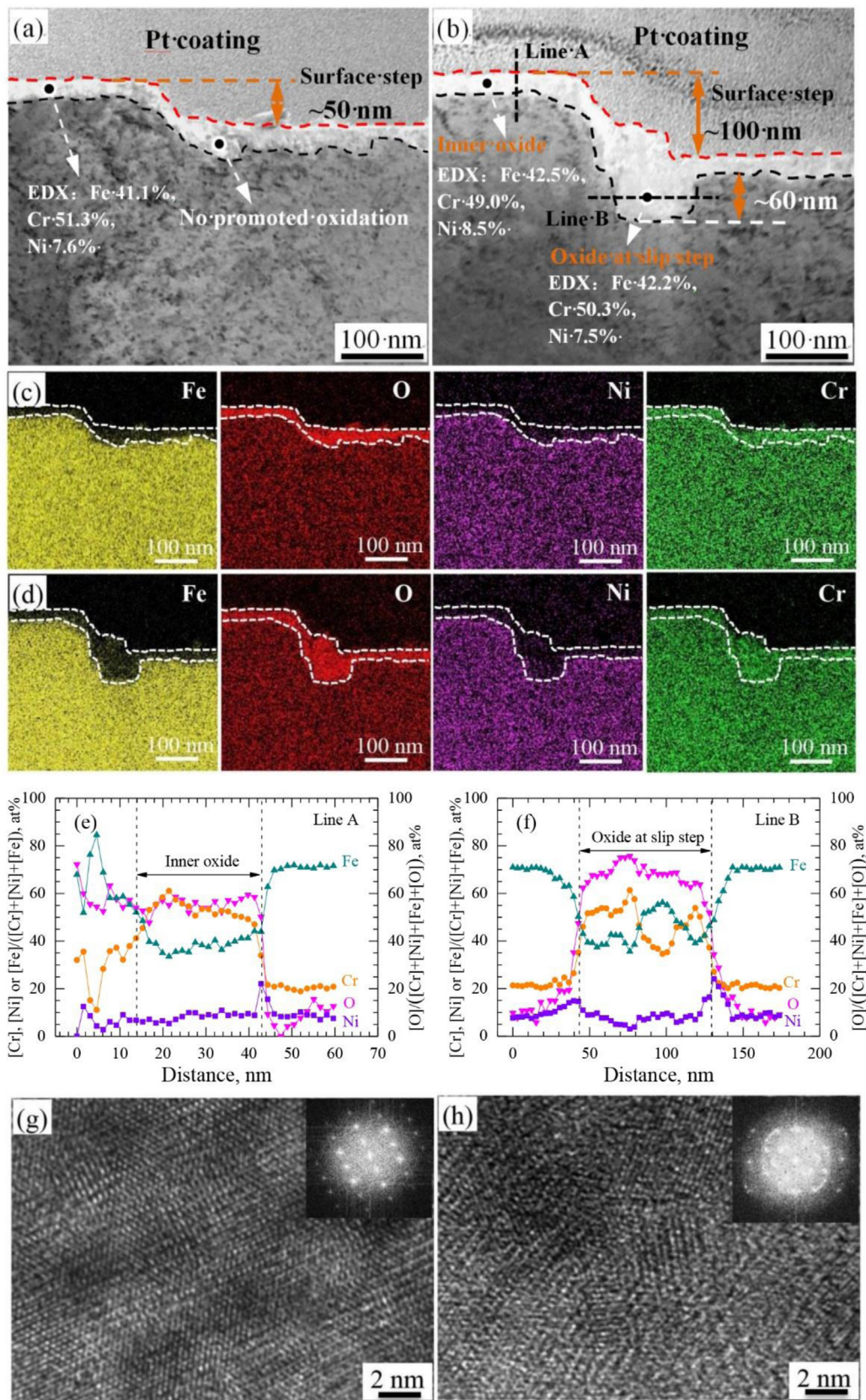


Fig. 5. (a) and (b) TEM observation of the cross-section of the oxide scale formed at two slip steps with a different height in 0.5-dpa irradiated 304NG SS. (c) and (d) The corresponding EDX mappings for Cr, Fe, Ni, and O showing oxidation at surface and along the slip steps. (e) and (f) EDX point-scans collected along lines A and B shown in (b), respectively. (g) and (h) High resolution TEM observation of the inner and the oxide formed at the slip step shown in (b).

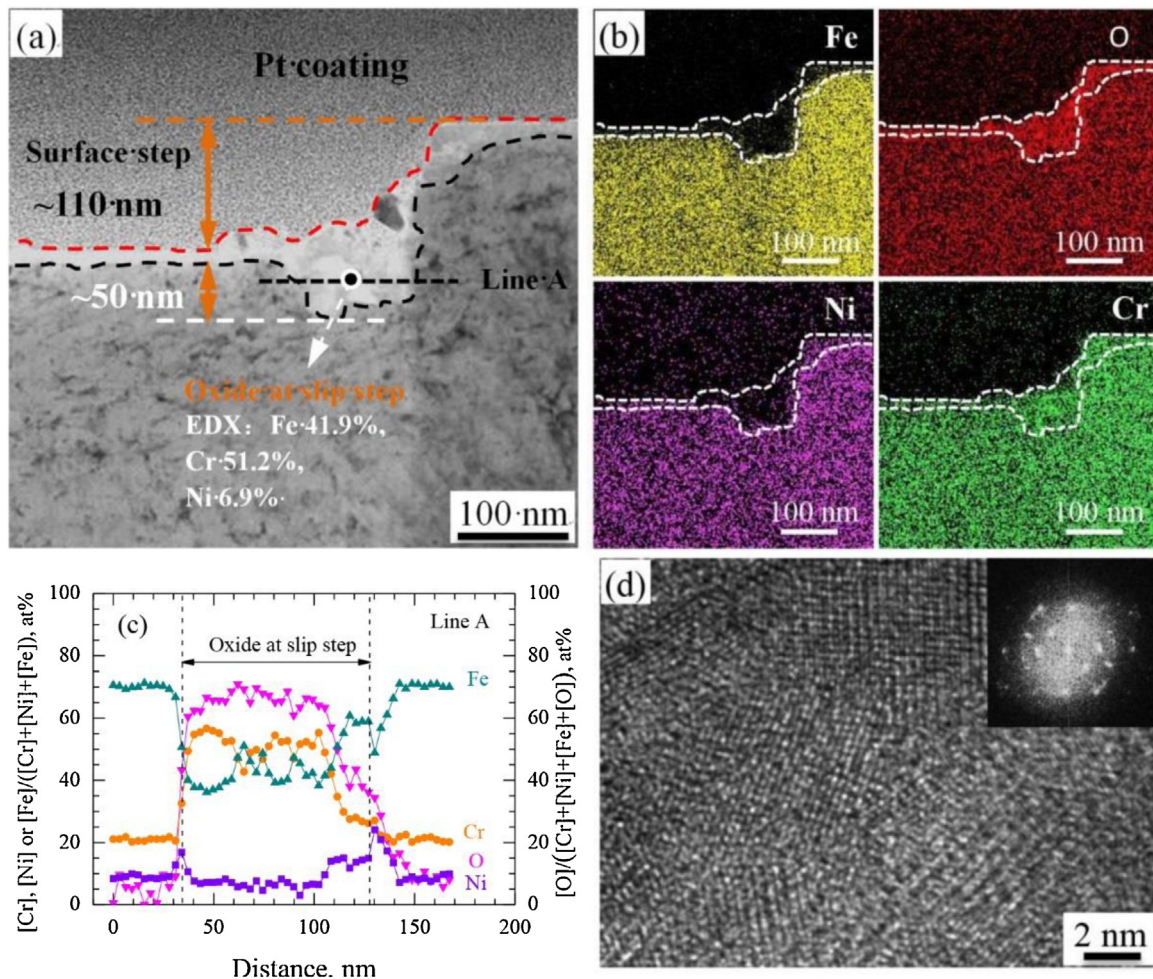


Fig. 6. (a) TEM observation of the localized corrosion at a slip step in 5-dpa irradiated 304NG SS, (b) the corresponding EDX mappings for Cr, Fe, Ni, and O showing oxidation along the slip step. (c) EDX point-scans collected along line A shown in (a). (d) High resolution TEM observation of the oxide formed at the slip step shown in (a).

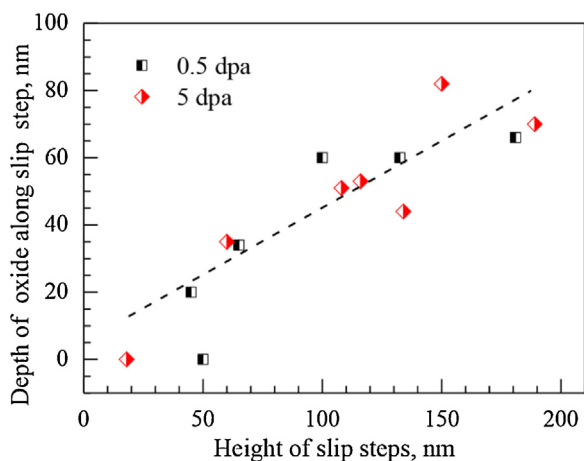


Fig. 7. Depth of oxide along the slip step as a function of slip step height.

2.5. Analyses of the IASCC behavior

At each interrupted strain, the IASCC test was temporarily stopped and the specimens were removed from the autoclave for a characterization of the IASCC behavior. Following the ultrasonic cleaning in pure water and ethanol, the gage surface of each specimen was observed using the FEI XL30 FEG-SEM for counting the

number of cracks and measuring the crack length over the irradiated region. Since the cracks only initiated in the irradiated region, the average crack length shown in the following equation can be defined as the total length of observed cracks per unit area:

$$\beta = L/S \quad (2)$$

where β is the crack length density, L is the total length of observed cracks in the irradiated region and S is the irradiated area. It is worth noting that only the cracks in the range of 10100 μm were selected for the statistics of crack number and average crack length [37].

3. Results

3.1. Localized deformation in irradiated 304NG SS

3.1.1. Weighted height of the slip step

The morphology and height of the slip step formed on the SA and irradiated specimens following the straining in the primary water as analyzed by AFM are shown in Fig. 2(a)–(f). The slip step observed was stripe in the surface and convex edge in the three-dimensional morphology. While the morphology of the slip step was similar, the height calculated by force-profile across the slip step showed difference, as displayed in Fig. 2(e) and (f). Fig. 3 shows a comparison of weighted average height of the slip step formed on the SA and irradiated specimens strained to 1% and 3% in the primary water. As can be seen, despite the large step-to-step variation in the slip step height, both the irradiation dose and applied

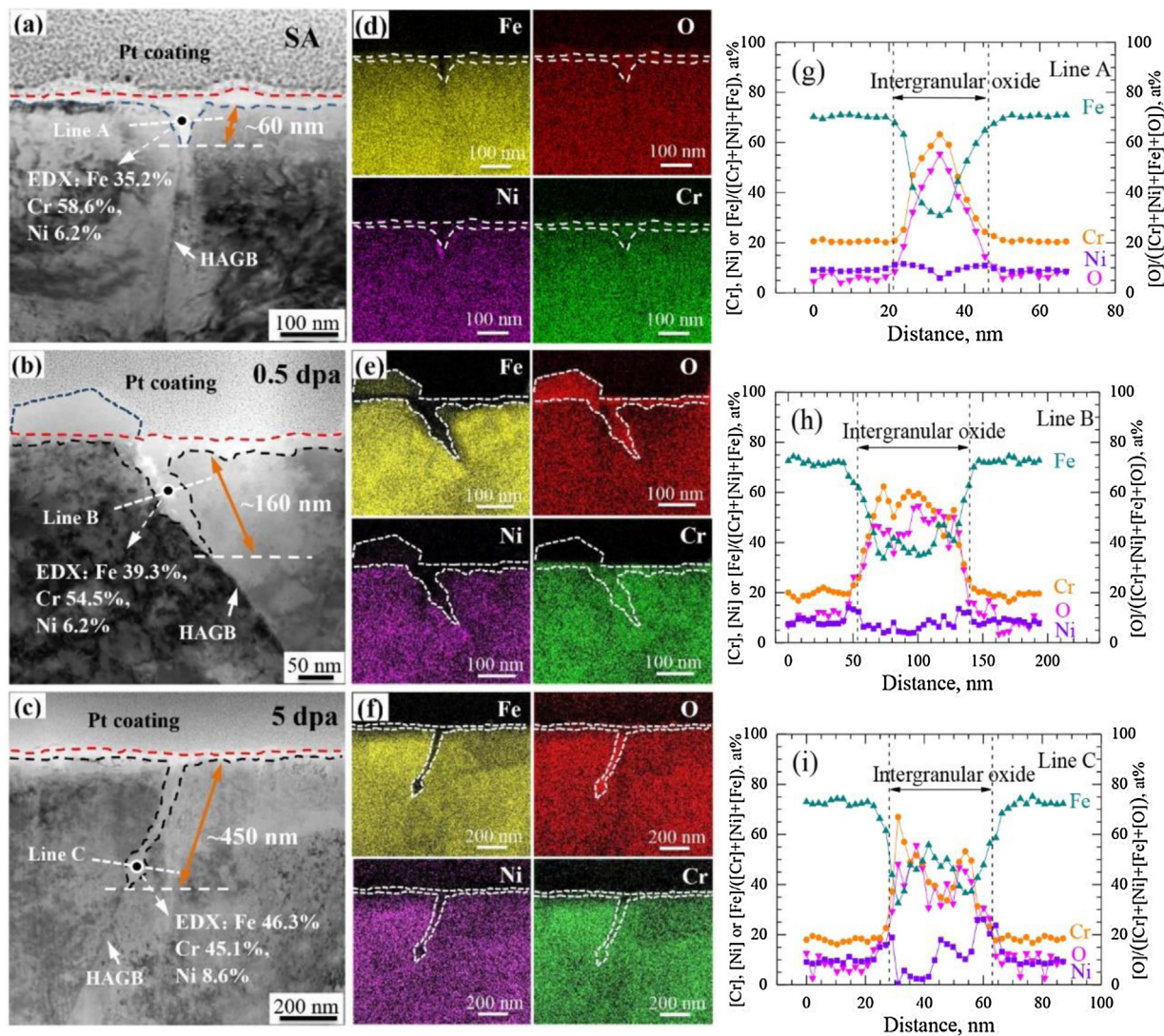


Fig. 8. TEM observation of the localized corrosion at a grain boundary in SA (a), 0.5- (b) and 5-dpa (c) irradiated 304NG SS. (d), (e) and (f) the corresponding EDX mappings for Cr, Fe, Ni, and O showing oxidation along the grain boundary shown in (a), (b) and (c), respectively. (g), (h) and (i) EDX point-scans collected along lines A, B and C shown in (a), (b) and (c), respectively.

strain promoted the weighted average slip step height. In particular, the weighted average slip step height was always larger at 5 dpa for the same strain level, indicating that increasing the irradiation dose led to a higher deformation heterogeneity of the irradiated steel.

3.1.2. Localized residual strain at grain boundary and slip step

The typical SEM morphology and color-coded M_L distribution mappings of the 0.5- and 5-dpa irradiated specimens following the 3% straining in the primary water are shown in Fig. 4. As illustrated, the residual strain developed heterogeneously in the irradiated steel due to the applied strain. In particular, the strain tended to accumulate to slip steps and grain boundaries, suggesting a strain concentration at these regions. The level of residual strain concentration at the grain boundary was increased by the irradiation dose (Fig. 4(a) and (b)). In addition, the 5-dpa irradiated specimen also exhibited a higher strain concentration level than the 0.5-dpa irradiated specimen at the slip step. This implies a correlation between the strain concentration and the height of the slip step, which is in agreement with the AFM analysis.

Fig. 4(c) shows the statistical distribution of the local misorientations with a regression curve optimized using the log-normal

distribution. The M_L distribution is seen to be well represented by the log-normal distribution, while the M_L for frequency peak in 5-dpa irradiated specimen was higher than that in 0.5-dpa irradiated specimen. This suggested that increasing irradiation dose increased the mean value of M_L and thus the strain concentration at the slip step and grain boundary.

3.2. Localized corrosion in irradiated 304NG SS in simulated primary PWR water

3.2.1. Localized corrosion at slip step

Fig. 5 shows the TEM observation and analysis of the cross-section of the oxide scale formed at two typical slip steps with a different height in 0.5-dpa irradiated specimen. It was found that the morphology of the oxide scale differed with the slip step height, shown in Fig. 5(a) and (b). A flat inner oxide film was observed to form at the slip step with a height of 50 nm, while the slip step with a larger height of 100 nm showed a penetration of oxidation with a depth of ~50 nm through the surface to the matrix, as displayed by the EDX mappings of Cr, Fe, Ni and O in Fig. 5(c) and (d). This suggested an acceleration of the localized corrosion by a larger slip step height. Both the EDX mappings and line scans revealed that the

composition of the inner oxide and the oxide formed at the slip step was similar (Fig. 5(e) and (f)). According to the high-resolution TEM as shown in Fig. 5(g) and (h), the oxides consisted of Fe-Cr-Ni spinel oxide [38,39]. In addition, there is no obvious difference in Cr concentration of the inner oxide and the oxide formed at the slip step in the 0.5-dpa irradiated specimens. Nevertheless, further observation and analysis by high-resolution TEM (Fig. 5(g) and (h)) also displayed a difference in structure of the inner oxide and the oxide formed at the slip step. The crystallographic pattern analyzed by FFT shown in the insets in the figures displayed a clear lattice character of the inner oxide, suggesting that the inner oxide was dominated by crystalline. On the other hand, the pattern of the oxide formed at the slip step showed a mixed character of polycrystalline diffraction rings and speckled diffraction lattice, suggesting a finer oxide grain in the oxide [38,40].

The localized corrosion at another slip step with a similar height of 110 nm in 5-dpa irradiated specimen was also observed and analyzed by TEM, shown in Fig. 6. The depth and microstructure of the oxide formed at the slip step were both similar to that in the 0.5-dpa irradiated specimen. This result as well as TEM observation shown in Fig. 5 suggested that the localized corrosion at the slip step was most likely dominated by the height of the slip step.

Fig. 7 summarized depth of the oxide formed at the slip step in 0.5- and 5-dpa irradiated specimens following the 3% straining. It showed that increasing the height promoted the localized corrosion along the slip step in irradiated 304NG SS in primary PWR water. This was in association with an increased deformation/strain at the slip step with a larger height.

3.2.2. Localized corrosion at grain boundary

The morphology and microstructure of the oxide formed at the grain boundary in SA, 0.5- and 5-dpa irradiated specimens following the 3% straining are shown in Fig. 8. A sharp oxide formed at the grain boundary was observed, suggesting a typical nature of intergranular corrosion. While the morphology of the oxide at the grain boundary was similar, the depth of the oxide showed difference in the specimens. In SA specimen, the oxide depth was approximately 60 nm, shown in Fig. 8(a). The oxide depth in 0.5- and 5-dpa irradiated specimens, however, was increased to ~160 nm and ~450 nm, respectively (shown in Fig. 8(b) and (c)).

EDX mappings and line-scans shown in Fig. 8(d)–(i) revealed the composition profiles of the oxide formed at the grain boundaries in SA, 0.5- and 5-dpa irradiated specimens. While all the oxides were depleted in Fe and enriched in Cr, the EDX point-scan analysis showed that the Cr content of the oxide at grain boundary was decreased by a higher irradiation dose of the specimen. This is attributed to the depletion of Cr at the primary grain boundary by the radiation induced segregation (RIS), as has been discussed in a previous study [22].

Fig. 9(a) summarized depth of the oxide formed at the grain boundary in SA, 0.5- and 5-dpa irradiated specimens following the 3% straining. A variation in oxidation depth was observed, which varied from ~80 to ~450 nm in the 5-dpa irradiated specimen, and from ~60 to ~160 nm in the SA and 0.5-dpa irradiated specimens. The variation was most likely due to the difference in the grain orientation with the tensile direction, which caused the difference in the localized deformation at the grain boundary. Nevertheless, the increase of the average oxidation depth by a higher irradiation dose was confirmed, as summarized in Fig. 9(b). Further, data reported in a previous study [22] on the oxidation depth at the grain boundary in strain-free specimens was also shown in Fig. 9(a) and (b) for a comparison. As can be seen, the increase of the oxidation depth with the irradiation dose was much higher in the 3% strained specimens, although the exposure time for the 3% strained specimens

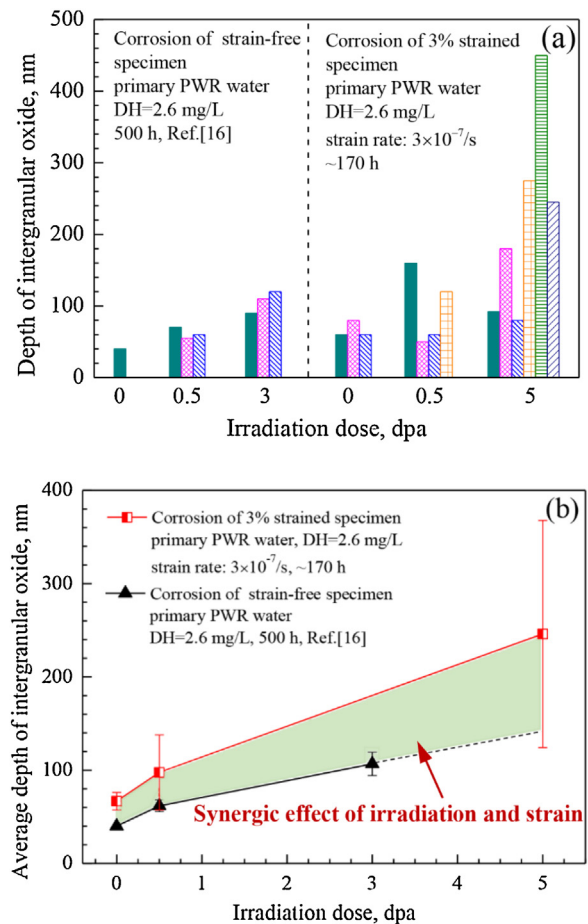


Fig. 9. (a) Depth of intergranular oxide in strain-free and 3% strained specimens as a function of irradiation dose. Each of the bars in the dose columns represents an analyzed grain boundary at the corresponding dose level. (b) Dependence of average depth of intergranular oxidation on irradiation dose.

was shorter than that for the strain-free specimens (~170 h for strained and 500 h for strain-free specimens). As shown in Fig. 9(b), the average oxidation depth in 0.5-dpa irradiated specimen was ~50 nm in the strain-free specimen, which was increased to ~95 nm by the 3% applied strain, and further to ~220 nm by increasing the irradiation to 5 dpa at the same level of applied strain. These results suggested a synergic effect of irradiation and strain on the intergranular corrosion of the irradiated steel.

3.3. IASCC in 304NG SS in simulated primary PWR water

The SEM observation of the cracking behavior on surfaces of the SA and irradiated specimens following the 1% and 3% straining is shown in Fig. 10. Typical intergranular cracking and deformation induced slip steps were observed on the irradiated specimens following the straining, while the SA specimen showed no such cracks but only a few slip steps. It was noting that intragranular cracking was also observed at the slip steps in the irradiated specimens, as shown in Fig. 11. Further, an observation of the 5-dpa irradiated specimen over the same region as shown in Fig. 10(g) and (h) revealed that the corroded and partly cracked grain boundaries at 1% strain were fully separated by an increased strain of 3%, suggesting a promotion effect of strain on cracking. Fig. 12 summarizes the observation results, which shows that both the number and length of the intergranular cracks were increased by a higher irradiation dose and applied strain.

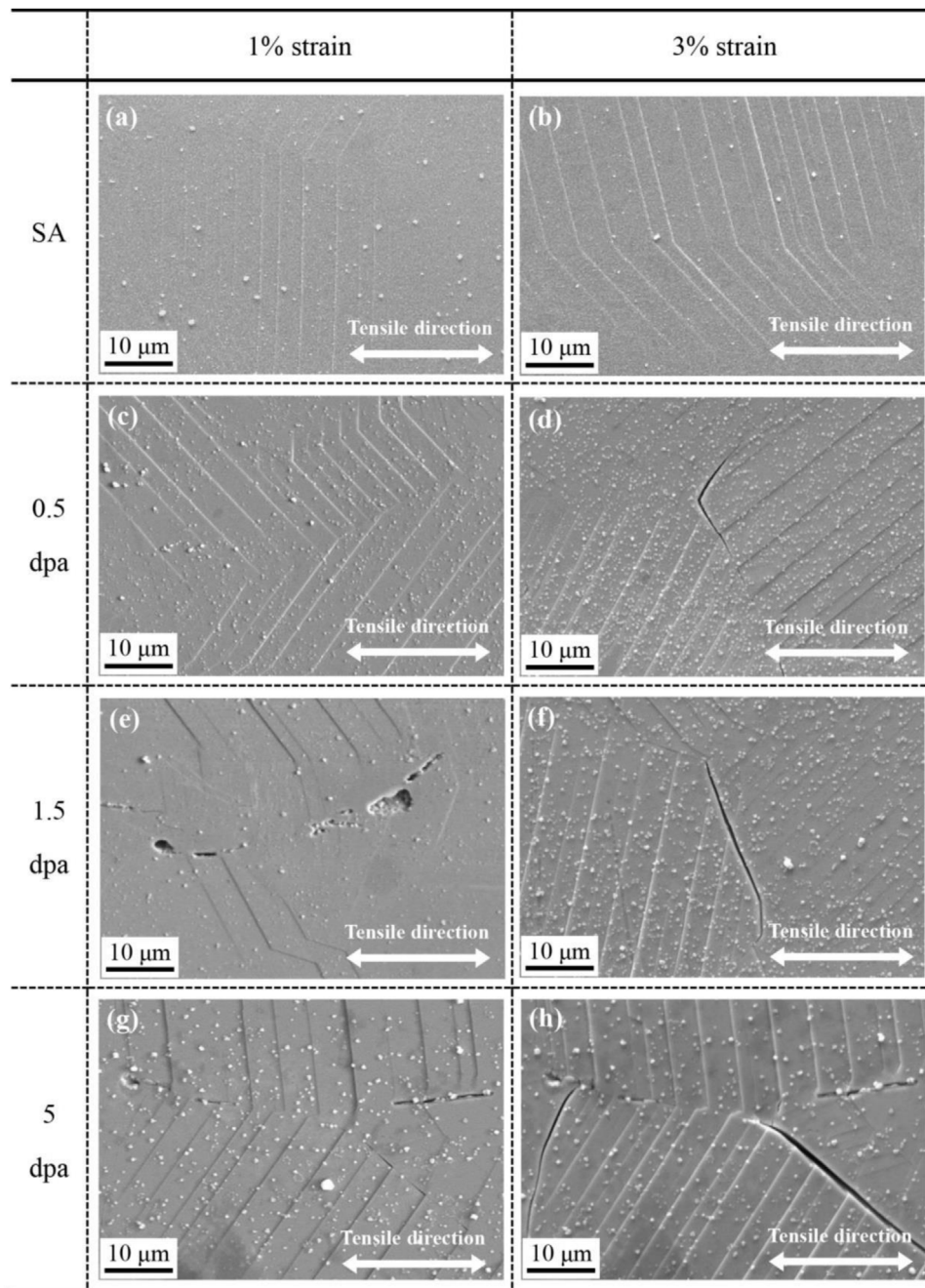


Fig. 10. SEM observation of surface cracking of 304NG SS subjected to different irradiation doses following a strain of 1% (left) and 3% (right) in simulated primary PWR water at 320 °C. (a, b) solution annealed, (c, d) 0.5 dpa, (e, f) 1.5 dpa, (g, h) 5 dpa. (g) and (h) were imaged from the same location for the purpose of the correlation between the applied strain and cracking initiation.

4. Discussion

4.1. Effect of irradiation on localized deformation of 304NG SS in primary PWR water

The above description of AFM and EBSD analysis showed that the height of slip steps, as well as the strain concentration at the slip step and grain boundary in the 5-dpa irradiated specimen was larger than those in the 0.5-dpa irradiated specimen. These results indicated that increasing irradiation dose led to a higher level of localized deformation at slip step and grain boundary in 304NG SS.

The effect of irradiation on localized deformation has been recognized in multiple irradiated austenitic alloys [6–8]. The increase

in slip step height for irradiated 304NG SS with irradiation dose is further confirmed in this study as shown in Fig. 3. At a higher irradiation dose, more dislocations induced by the irradiation are accumulated to mitigate homogeneous deformation. As such more deformation was localized into the dislocation channel, leading to a higher slip step height [7,41,42]. On the other hand, since the slip step transmits or terminates at the grain boundary as shown in Fig. 10, it will eventually impact on the grain boundary and promote localized grain boundary deformation [7,10,12]. As a result, the localized deformation in 5-dpa irradiated specimen is larger than that in 0.5-dpa irradiated specimen because a higher slip step is likely to have a stronger impact on the grain boundary, leading to a higher level of strain concentration at the grain boundary as illustrated in Fig. 4.

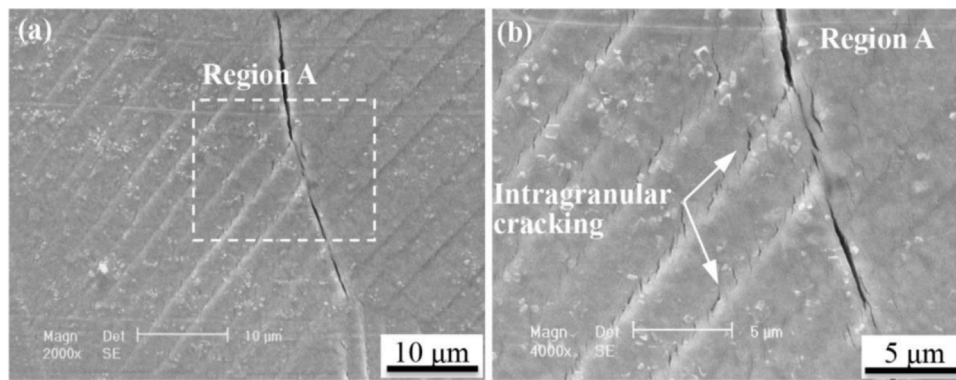


Fig. 11. (a) SEM observation of intragranular cracking at slip step in 1.5-dpa irradiated 304NG SS following a strain of 3% in simulated primary PWR water at 320 °C. (b) High magnification SEM images of the selected region A in (a).

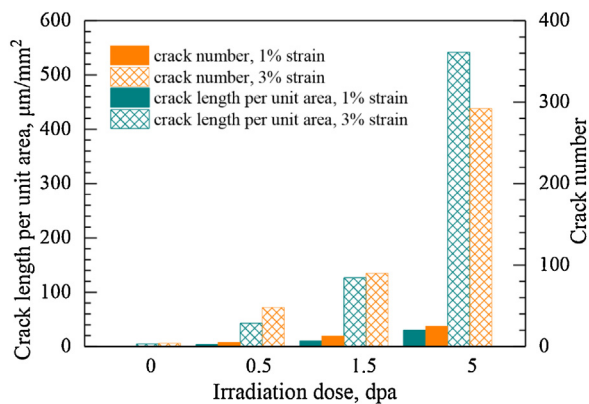


Fig. 12. IASCC susceptibility as measured by crack number and crack length per unit area as a function of irradiation dose.

4.2. Effect of localized deformation on localized corrosion of 304NG SS

The TEM observation and analysis shown in Figs. 5 and 8 revealed that increasing irradiation dose promoted localized corrosion along the slip step and grain boundary. Further, as discussed above, increasing the irradiation dose also led to a higher localized deformation at the slip step and grain boundary. This indicated a correlation between the enhanced corrosion and localized deformation.

The higher slip step corresponds to a highly deformed microstructure, in which enhanced corrosion could occur easily by the process of stress/strain assisted oxidation [30,43]. This is in accordance with the enhanced localized corrosion along deformation band in cold worked steel [25,26]. Further, according to the high resolution TEM analysis shown in Fig. 5(g) and (h), the grain size of the oxide formed at the slip step is much smaller than that of the inner oxide. This also suggested a higher localized corrosion rate along the slip step, since the finer grain of the oxide could certainly promote the oxygen diffusion by providing more diffusion paths [44–46]. The formation of polycrystalline oxide could relate to the continuing changing of step height during deformation and therefore induce breaking of the oxide and formation of small oxides.

The above discussion suggests that the localized deformation assisted diffusion of oxygen is the primary cause for enhanced localized corrosion along the slip step. The deformation, on the other hand, also promotes the intergranular corrosion by means of strain concentration at the grain boundary. During the exposure in the primary water, a homogeneous and flat oxide was formed at first

on the surfaces of the 0.5- and 5-dpa irradiated specimens. Then enhanced intergranular corrosion occurred due to Cr-depletion induced by RIS at the grain boundary, as has been acknowledged in a previous study [22]. In the following process of straining, the intergranular corrosion is related to the combined effect of RIS and strain concentration at the grain boundary. Further, the strain concentration at the grain boundary in 5-dpa irradiated specimen is heavier than that in 0.5-dpa irradiated specimen, as discussed previously in section 4.1. As concluded in other studies [26,30,43,47], strain concentration could accelerate the oxygen diffusion along grain boundaries, causing a further grain boundary oxidation.

4.3. Correlation of localized deformation and localized corrosion with IASCC

The SEM observation and analysis shown in Figs. 10 and 11 revealed that increasing the irradiation dose resulted in a higher IASCC susceptibility of 304NG SS. Further, the discussion in Sections 4.1 and 4.2 suggested that increasing the irradiation dose also promoted the localized corrosion along the slip step and grain boundary, in correspondence with a localized residual strain concentration in these regions. This indicated a correlation between the higher IASCC susceptibility and the enhanced localized deformation and corrosion along the slip step and grain boundary.

As has been acknowledged, the residual strain concentration at grain boundary in austenitic alloys could promote the cracking susceptibility in high temperature water [27,28,48–51]. As mentioned previously, straining could result in formation of a higher slip step and heavier lattice distortion at the grain boundary in the specimen subjected to a higher irradiation dose. This was correlated with the deformation heterogeneity and well described by the number of dislocations that were localized into the dislocation channel and accumulated to the grain boundary [7,8]. The slip step and distorted grain boundary indicated high dislocation density and strain concentration (Fig. 4), which accelerated the cracking initiation.

The strain concentration, on the other hand, affected the localized corrosion behavior at the slip step and grain boundary. As the oxidation under stress/strain favors the formation of cracking initiation [32,33,43,52], a higher corrosion rate at the slip step and grain boundary in the steel with a higher irradiation dose caused a higher cracking susceptibility, shown in Figs. 10 and 11. The oxide at the grain boundary promoted the initiation of a typical intergranular SCC, while the oxide at slip step promoted the intragranular cracking initiation since most part of the slip step was within the grain. As such, increasing the irradiation dose promoted SCC initiation due to the enhanced localized corrosion at the grain boundary and slip step. Further, as a higher irradiation dose led to generation of more slip steps with a higher height (Figs. 2 and 3), increasing the

irradiation dose would favor formation of sites for SCC initiation as a result of the increased localized corrosion along the slip step.

5. Conclusions

Irradiation assisted localized corrosion and stress corrosion cracking in 304NG SS in simulated primary PWR water were investigated. The following conclusions can be drawn from the investigation:

- (1) Increasing the irradiation dose led to a higher localized deformation at grain boundary and slip step of 304NG SS due to the irradiation induced defects.
- (2) Irradiation and strain promote intergranular corrosion of 304NG SS in simulated PWR primary water. This is attributed to the synergic effect of irradiation and strain, which resulted in a strain concentration and a depletion of Cr at the grain boundary by RIS.
- (3) The depth of the oxidation at slip step of irradiated 304NG SS in primary PWR water is increased by a higher height of the slip step due to the accumulated localized strain by a higher irradiation dose.
- (4) Increasing irradiation dose within the studied dose range (0.55 dpa) promoted IASCC initiation as a result of the synergic effect of the enhanced localized deformation and corrosion at the grain boundary and slip step.

Acknowledgements

This work was financially supported by the International Science & Technology Cooperation Program of China (No. 2014DFA50800), and partly supported by the Essential Research Fund by SNPTC (No. 2015SN010-007). One author, Qunjia Peng, would thank the support of Natural Science Foundation of Jiangsu Province (BK20191178). The authors gratefully acknowledge the Michigan Ion Beam Laboratory at the University of Michigan for performing proton irradiation experiments.

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